

There were 3 people that participated in CSI data collection. In this report I call it one session for each person.

Using the strategy to reduce the patience reduced model complexity and increased regularization and balanced class weights from training data.

I used this configuration as follows:

Convolutional Layer:

- Filters: 32
- Kernel Size: 3
- Activation Function: ReLU ([relu](#))

LSTM Layer:

- Units: 64

Dense Hidden Layer:

- Units: 64
- Activation Function: ReLU ([relu](#))

Output Layer:

- Activation Function: Softmax ([softmax](#))

Regularization:

- First Dropout Rate: 0.3 (30%)
- Second Dropout Rate: 0.5 (50%)

Compilation:

- Optimizer: Adam ([adam](#))
- Loss Function: Sparse Categorical Crossentropy ([sparse\\_categorical\\_crossentropy](#))
- Metric: Accuracy ([accuracy](#))

# Same Session

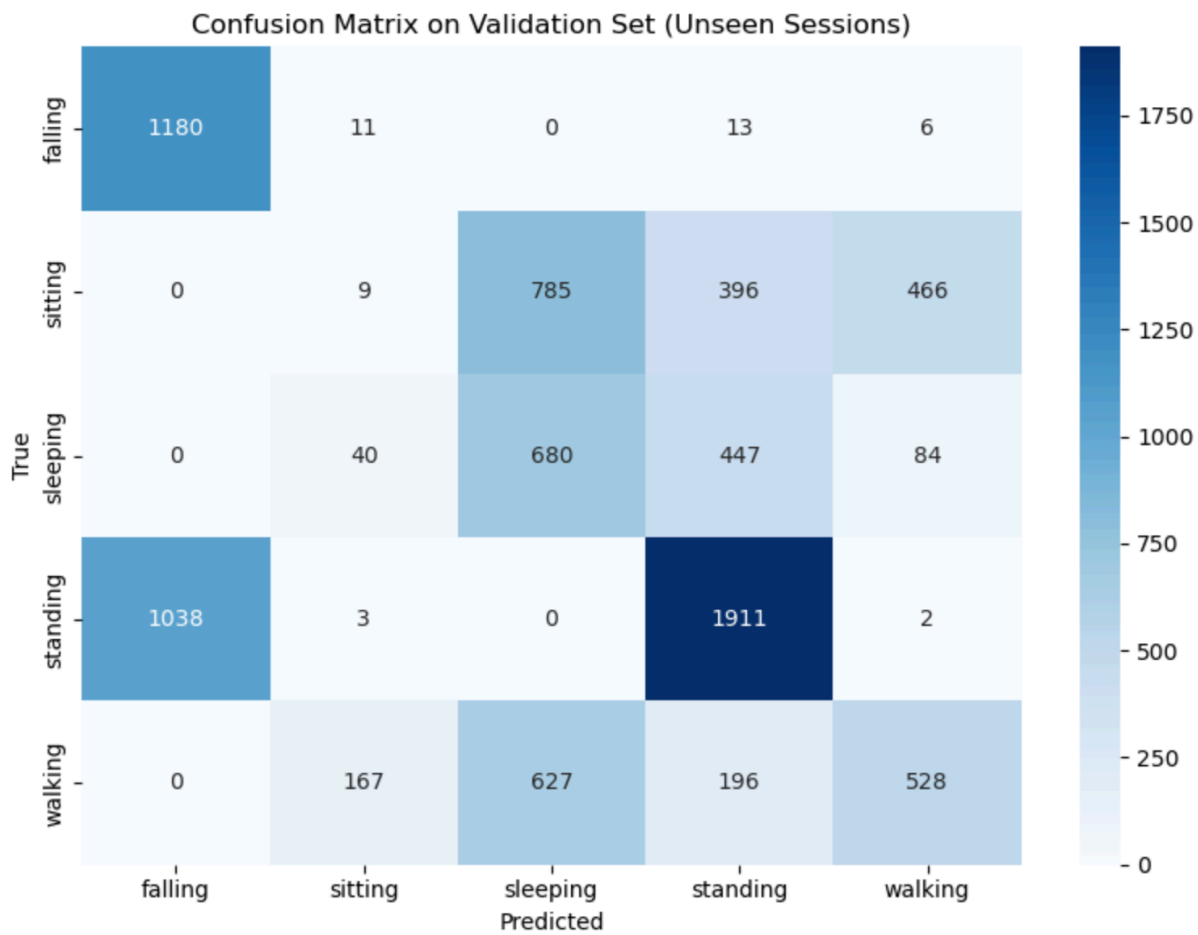
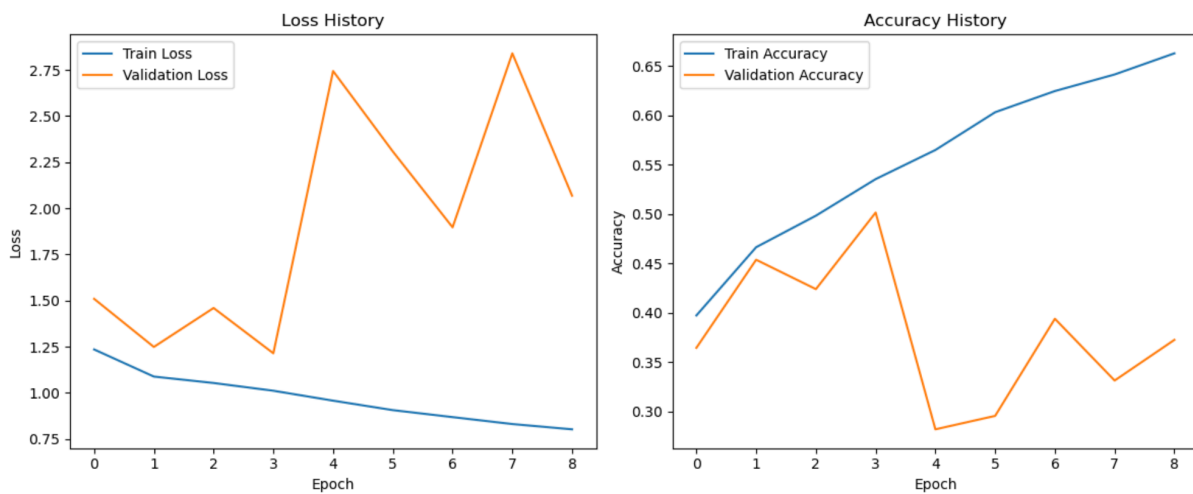
## Session 1

```
train_sessions_files = [  
    'esp32_csi_walking_20250727_152811',  
    'esp32_csi_walking_20250727_152811',  
    'esp32_csi_walking_20250727_152811',  
    'esp32_csi_standing_20250727_155036',  
    'esp32_csi_standing_20250727_155036',  
    'esp32_csi_standing_20250727_154516',  
    'esp32_csi_sleeping_20250727_164127',  
    'esp32_csi_sleeping_20250727_163829',  
    'esp32_csi_sleeping_20250727_163620',  
    'esp32_csi_sitting_20250727_161517',  
    'esp32_csi_sitting_20250727_161310',  
    'esp32_csi_sitting_20250727_160929',  
    'esp32_csi_falling_20250727_133043',  
    'esp32_csi_falling_20250727_132826',  
    'esp32_csi_falling_20250727_132620'  
]
```

```
val_sessions_files = [  
    'esp32_csi_walking_20250727_150913',  
    'esp32_csi_standing_20250707_155353',  
    'esp32_csi_sleeping_20250727_161957',  
    'esp32_csi_sitting_20250727_155647',  
    'esp32_csi_falling_20250727_131141'  
]
```

## Training Logs and Results:

```
Starting model training...  
Epoch 1/50  
626/626 - 21s - 34ms/step - accuracy: 0.3972 - loss: 1.2356 - val_accuracy: 0.3644 - val_loss: 1.5099  
Epoch 2/50  
626/626 - 12s - 20ms/step - accuracy: 0.4664 - loss: 1.0885 - val_accuracy: 0.4537 - val_loss: 1.2491  
Epoch 3/50  
626/626 - 12s - 20ms/step - accuracy: 0.4984 - loss: 1.0537 - val_accuracy: 0.4239 - val_loss: 1.4602  
Epoch 4/50  
626/626 - 13s - 20ms/step - accuracy: 0.5355 - loss: 1.0118 - val_accuracy: 0.5016 - val_loss: 1.2147  
Epoch 5/50  
626/626 - 12s - 20ms/step - accuracy: 0.5650 - loss: 0.9579 - val_accuracy: 0.2818 - val_loss: 2.7443  
Epoch 6/50  
626/626 - 13s - 21ms/step - accuracy: 0.6033 - loss: 0.9066 - val_accuracy: 0.2954 - val_loss: 2.3082  
Epoch 7/50  
626/626 - 13s - 21ms/step - accuracy: 0.6248 - loss: 0.8688 - val_accuracy: 0.3939 - val_loss: 1.8974  
Epoch 8/50  
626/626 - 13s - 20ms/step - accuracy: 0.6416 - loss: 0.8312 - val_accuracy: 0.3312 - val_loss: 2.8402  
Epoch 9/50  
626/626 - 13s - 21ms/step - accuracy: 0.6630 - loss: 0.8027 - val_accuracy: 0.3725 - val_loss: 2.0687
```

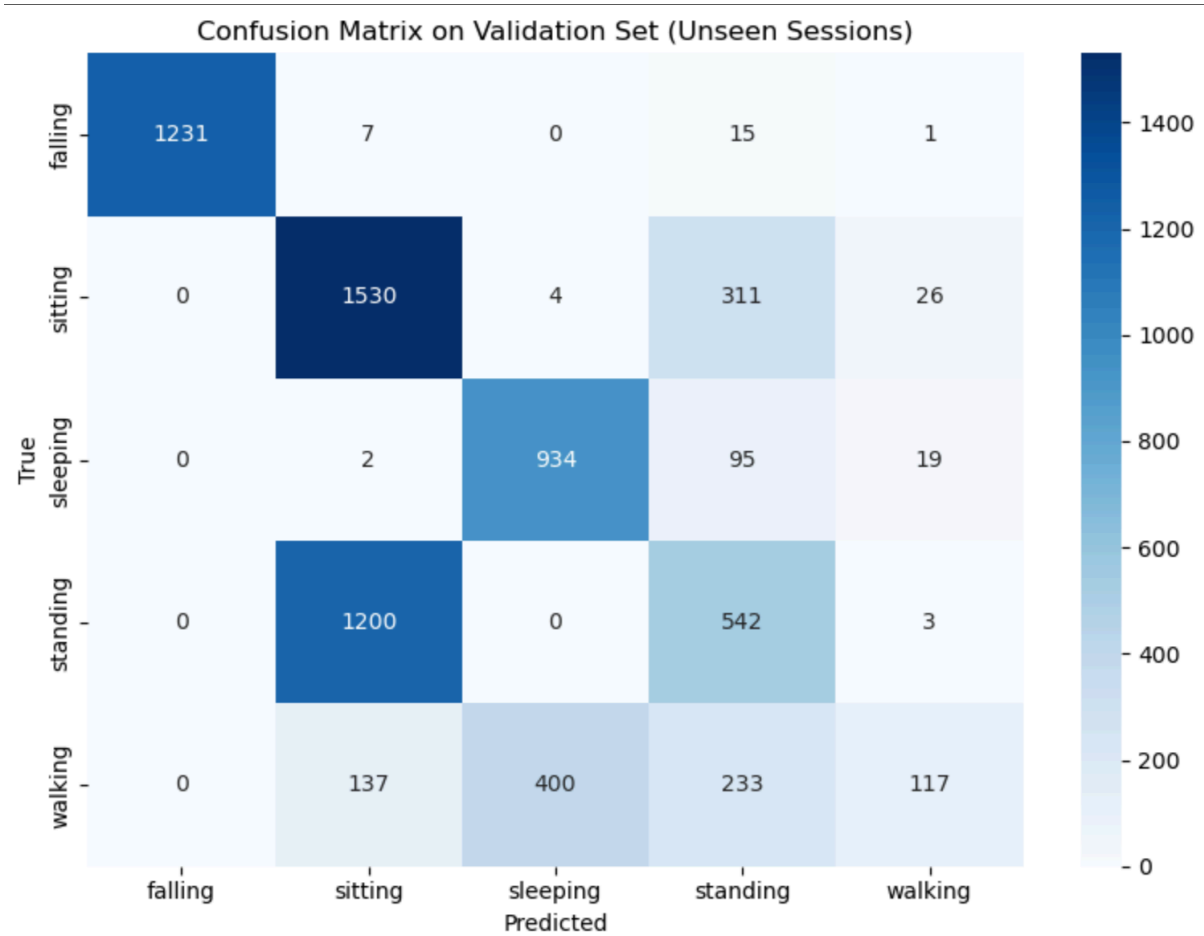
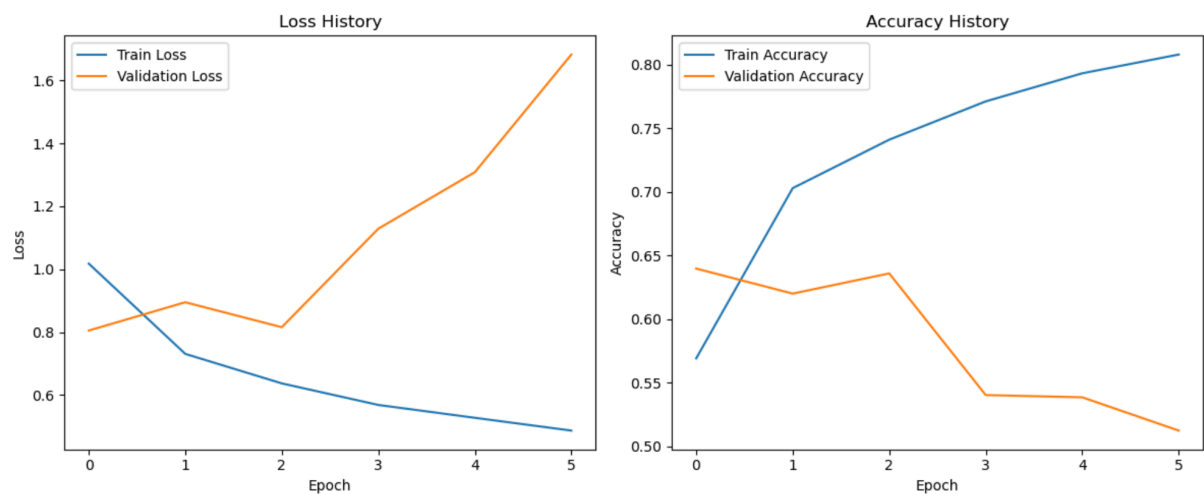


## Session 2 (Best)

```
train_sessions_files = [  
    'esp32_csi_falling_20250727_164609',  
    'esp32_csi_falling_20250727_164810',  
    'esp32_csi_falling_20250727_165221',  
    'esp32_csi_sitting_20250728_125533',  
    'esp32_csi_sitting_20250728_125737',  
    'esp32_csi_sitting_20250728_125939',  
    'esp32_csi_sleeping_20250728_115816',  
    'esp32_csi_sleeping_20250728_120022',  
    'esp32_csi_sleeping_20250728_120223',  
    'esp32_csi_standing_20250728_134435',  
    'esp32_csi_standing_20250728_134638',  
    'esp32_csi_standing_20250728_134840',  
    'esp32_csi_walking_20250728_143333',  
    'esp32_csi_walking_20250728_144158',  
    'esp32_csi_walking_20250728_145238'  
]  
  
val_sessions_files = [  
    'esp32_csi_falling_20250727_165423',  
    'esp32_csi_sitting_20250728_131337',  
    'esp32_csi_sleeping_20250728_121651',  
    'esp32_csi_standing_20250728_135733',  
    'esp32_csi_walking_20250728_144601'  
]
```

### Training Logs and Results:

```
Starting model training...  
Epoch 1/50  
614/614 - 22s - 35ms/step - accuracy: 0.5690 - loss: 1.0177 - val_accuracy: 0.6396 - val_loss: 0.8051  
Epoch 2/50  
614/614 - 13s - 21ms/step - accuracy: 0.7029 - loss: 0.7309 - val_accuracy: 0.6200 - val_loss: 0.8950  
Epoch 3/50  
614/614 - 13s - 21ms/step - accuracy: 0.7410 - loss: 0.6371 - val_accuracy: 0.6358 - val_loss: 0.8156  
Epoch 4/50  
614/614 - 13s - 20ms/step - accuracy: 0.7711 - loss: 0.5685 - val_accuracy: 0.5402 - val_loss: 1.1287  
Epoch 5/50  
614/614 - 13s - 21ms/step - accuracy: 0.7931 - loss: 0.5277 - val_accuracy: 0.5384 - val_loss: 1.3083  
Epoch 6/50  
614/614 - 13s - 21ms/step - accuracy: 0.8079 - loss: 0.4871 - val_accuracy: 0.5123 - val_loss: 1.6823
```



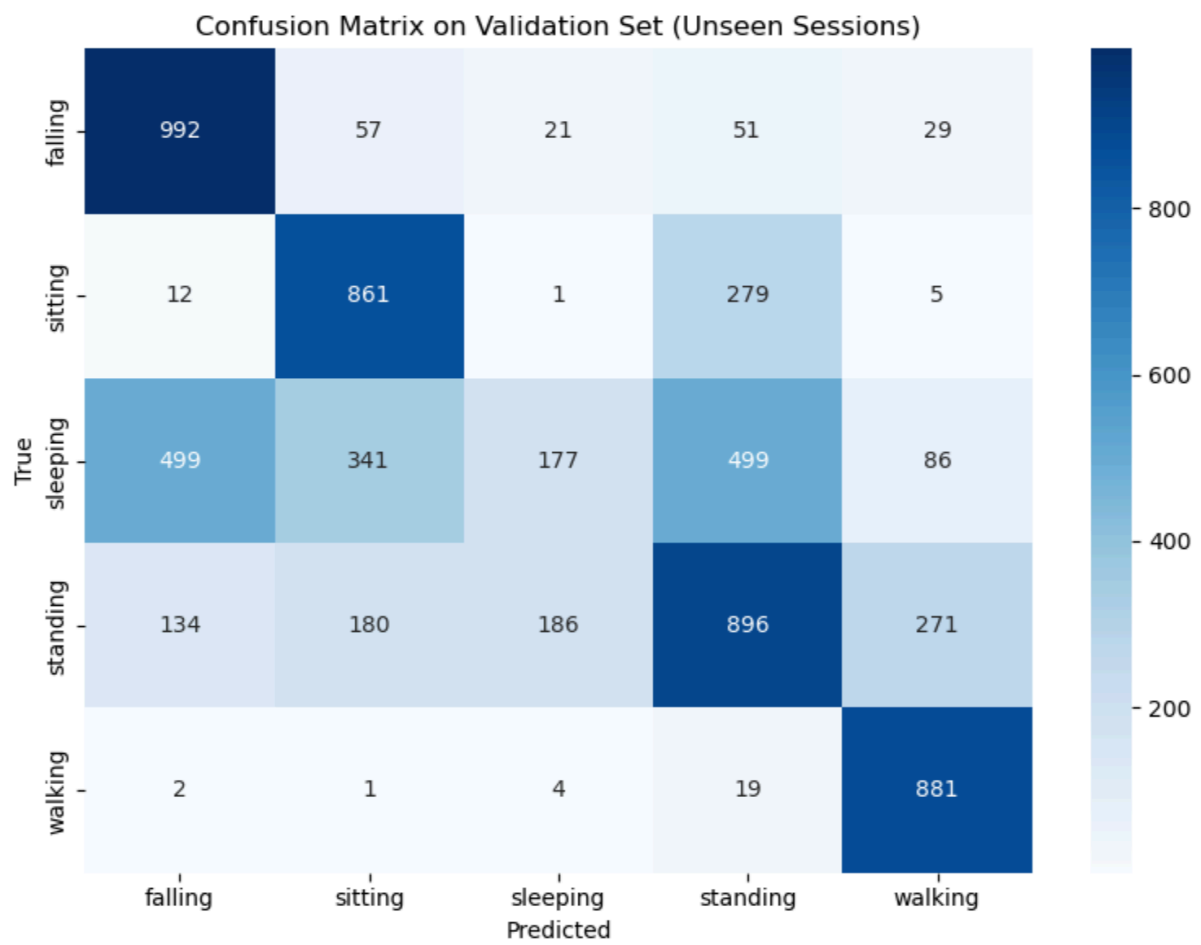
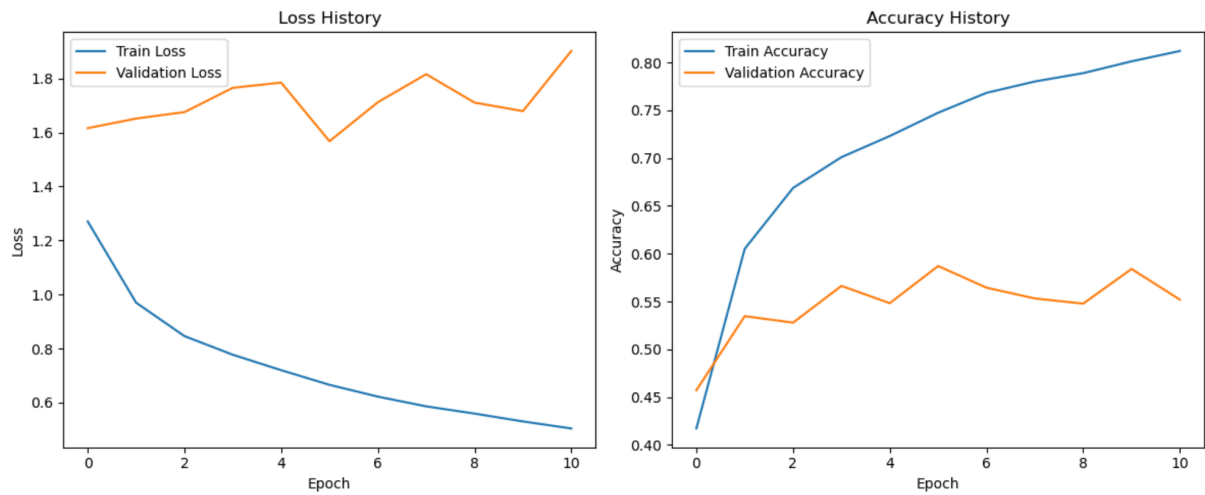
## Session 3

```
train_sessions_files = [  
    'esp32_csi_falling_20250728_113607',  
    'esp32_csi_falling_20250728_113809',  
    'esp32_csi_falling_20250728_114015',  
    'esp32_csi_sitting_20250728_131836',  
    'esp32_csi_sitting_20250728_132048',  
    'esp32_csi_sitting_20250728_132339',  
    'esp32_csi_sleeping_20250728_122101',  
    'esp32_csi_sleeping_20250728_122306',  
    'esp32_csi_sleeping_20250728_122512',  
    'esp32_csi_standing_20250728_140622',  
    'esp32_csi_standing_20250728_140838',  
    'esp32_csi_standing_20250728_141039',  
    'esp32_csi_walking_20250728_145517',  
    'esp32_csi_walking_20250728_145721',  
    'esp32_csi_walking_20250728_145927'  
]
```

```
val_sessions_files = [  
    'esp32_csi_falling_20250728_115449',  
    'esp32_csi_sitting_20250728_133848',  
    'esp32_csi_sleeping_20250728_125027',  
    'esp32_csi_standing_20250728_142621',  
    'esp32_csi_walking_20250728_152215'  
]
```

### Training Logs and Results:

```
Starting model training...  
Epoch 1/50  
608/608 - 21s - 35ms/step - accuracy: 0.4172 - loss: 1.2706 - val_accuracy: 0.4571 - val_loss: 1.6160  
Epoch 2/50  
608/608 - 12s - 21ms/step - accuracy: 0.6051 - loss: 0.9700 - val_accuracy: 0.5347 - val_loss: 1.6516  
Epoch 3/50  
608/608 - 13s - 21ms/step - accuracy: 0.6688 - loss: 0.8461 - val_accuracy: 0.5279 - val_loss: 1.6755  
Epoch 4/50  
608/608 - 12s - 21ms/step - accuracy: 0.7010 - loss: 0.7771 - val_accuracy: 0.5663 - val_loss: 1.7653  
Epoch 5/50  
608/608 - 12s - 20ms/step - accuracy: 0.7231 - loss: 0.7199 - val_accuracy: 0.5483 - val_loss: 1.7846  
Epoch 6/50  
608/608 - 13s - 21ms/step - accuracy: 0.7475 - loss: 0.6656 - val_accuracy: 0.5871 - val_loss: 1.5676  
Epoch 7/50  
608/608 - 13s - 21ms/step - accuracy: 0.7683 - loss: 0.6220 - val_accuracy: 0.5645 - val_loss: 1.7120  
Epoch 8/50  
608/608 - 13s - 21ms/step - accuracy: 0.7801 - loss: 0.5859 - val_accuracy: 0.5532 - val_loss: 1.8158  
Epoch 9/50  
608/608 - 13s - 21ms/step - accuracy: 0.7889 - loss: 0.5591 - val_accuracy: 0.5478 - val_loss: 1.7107  
Epoch 10/50  
608/608 - 13s - 21ms/step - accuracy: 0.8012 - loss: 0.5302 - val_accuracy: 0.5841 - val_loss: 1.6791  
Epoch 11/50  
608/608 - 13s - 21ms/step - accuracy: 0.8121 - loss: 0.5041 - val_accuracy: 0.5520 - val_loss: 1.9018
```



# Mix Sessions

## Session 1 as Validation

```
train_sessions_files = [  
    'esp32_csi_falling_20250728_113607',  
    'esp32_csi_falling_20250728_113809',  
    'esp32_csi_falling_20250727_165013',  
    'esp32_csi_sitting_20250728_131836',  
    'esp32_csi_sitting_20250728_132048',  
    'esp32_csi_sitting_20250728_130155',  
    'esp32_csi_sleeping_20250728_122101',  
    'esp32_csi_sleeping_20250728_122306',  
    'esp32_csi_sleeping_20250728_120022',  
    'esp32_csi_standing_20250728_140622',  
    'esp32_csi_standing_20250728_140838',  
    'esp32_csi_standing_20250728_134840',  
    'esp32_csi_walking_20250728_145517',  
    'esp32_csi_walking_20250728_145721',  
    'esp32_csi_walking_20250728_144158'  
]
```

```
val_sessions_files = [  
    'esp32_csi_walking_20250727_151943',  
    'esp32_csi_standing_20250727_155242',  
    'esp32_csi_sleeping_20250727_163416',  
    'esp32_csi_sitting_20250727_160513',  
    'esp32_csi_falling_20250727_132417',  
]
```

Training Logs and Results:

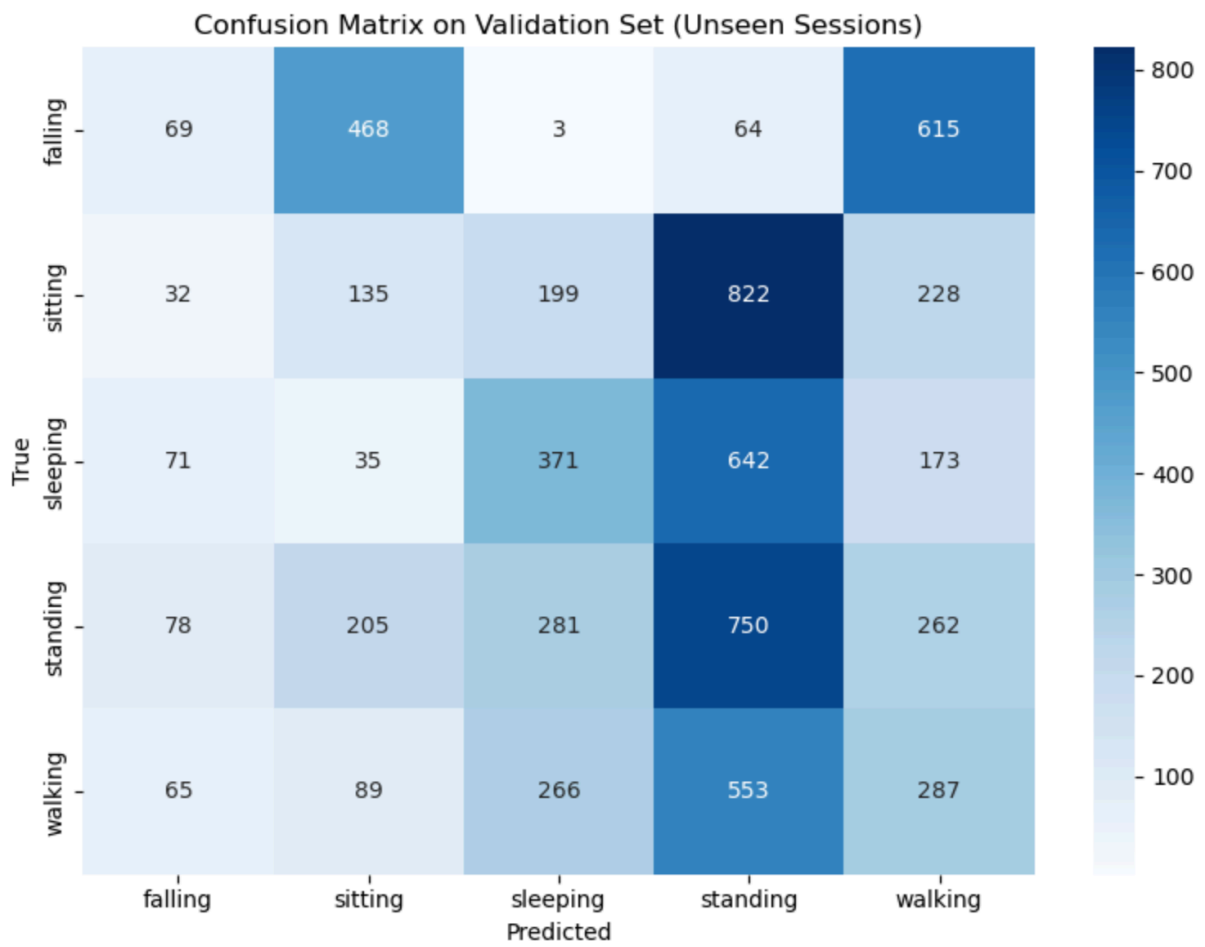


```

Starting model training...
Epoch 1/50
613/613 - 21s - 34ms/step - accuracy: 0.4123 - loss: 1.3328 - val_accuracy: 0.2554 - val_loss: 2.5989
Epoch 2/50
613/613 - 13s - 21ms/step - accuracy: 0.5854 - loss: 1.0259 - val_accuracy: 0.2514 - val_loss: 2.2946
Epoch 3/50
613/613 - 12s - 20ms/step - accuracy: 0.6431 - loss: 0.8953 - val_accuracy: 0.2410 - val_loss: 2.3256
Epoch 4/50
613/613 - 12s - 20ms/step - accuracy: 0.6686 - loss: 0.8330 - val_accuracy: 0.2336 - val_loss: 2.3389
Epoch 5/50
613/613 - 12s - 20ms/step - accuracy: 0.6877 - loss: 0.7990 - val_accuracy: 0.2339 - val_loss: 2.2842
Epoch 6/50
613/613 - 13s - 21ms/step - accuracy: 0.7027 - loss: 0.7602 - val_accuracy: 0.2366 - val_loss: 2.4408
Epoch 7/50
613/613 - 13s - 21ms/step - accuracy: 0.7230 - loss: 0.7219 - val_accuracy: 0.2503 - val_loss: 2.0853
Epoch 8/50
613/613 - 13s - 21ms/step - accuracy: 0.7316 - loss: 0.6971 - val_accuracy: 0.2296 - val_loss: 2.0763
Epoch 9/50
613/613 - 13s - 21ms/step - accuracy: 0.7444 - loss: 0.6652 - val_accuracy: 0.2128 - val_loss: 2.0679
Epoch 10/50
613/613 - 13s - 21ms/step - accuracy: 0.7585 - loss: 0.6399 - val_accuracy: 0.2372 - val_loss: 1.9757
Epoch 11/50
613/613 - 13s - 21ms/step - accuracy: 0.7652 - loss: 0.6155 - val_accuracy: 0.2384 - val_loss: 1.8406
Epoch 12/50
613/613 - 12s - 20ms/step - accuracy: 0.7828 - loss: 0.5890 - val_accuracy: 0.2378 - val_loss: 1.9572
Epoch 13/50
613/613 - 12s - 20ms/step - accuracy: 0.7896 - loss: 0.5698 - val_accuracy: 0.2110 - val_loss: 1.8903
Epoch 14/50
613/613 - 13s - 21ms/step - accuracy: 0.7951 - loss: 0.5541 - val_accuracy: 0.2135 - val_loss: 2.0786
Epoch 15/50
613/613 - 13s - 21ms/step - accuracy: 0.8081 - loss: 0.5317 - val_accuracy: 0.2444 - val_loss: 2.0036
Epoch 16/50
613/613 - 13s - 21ms/step - accuracy: 0.8104 - loss: 0.5187 - val_accuracy: 0.2318 - val_loss: 1.9518

```





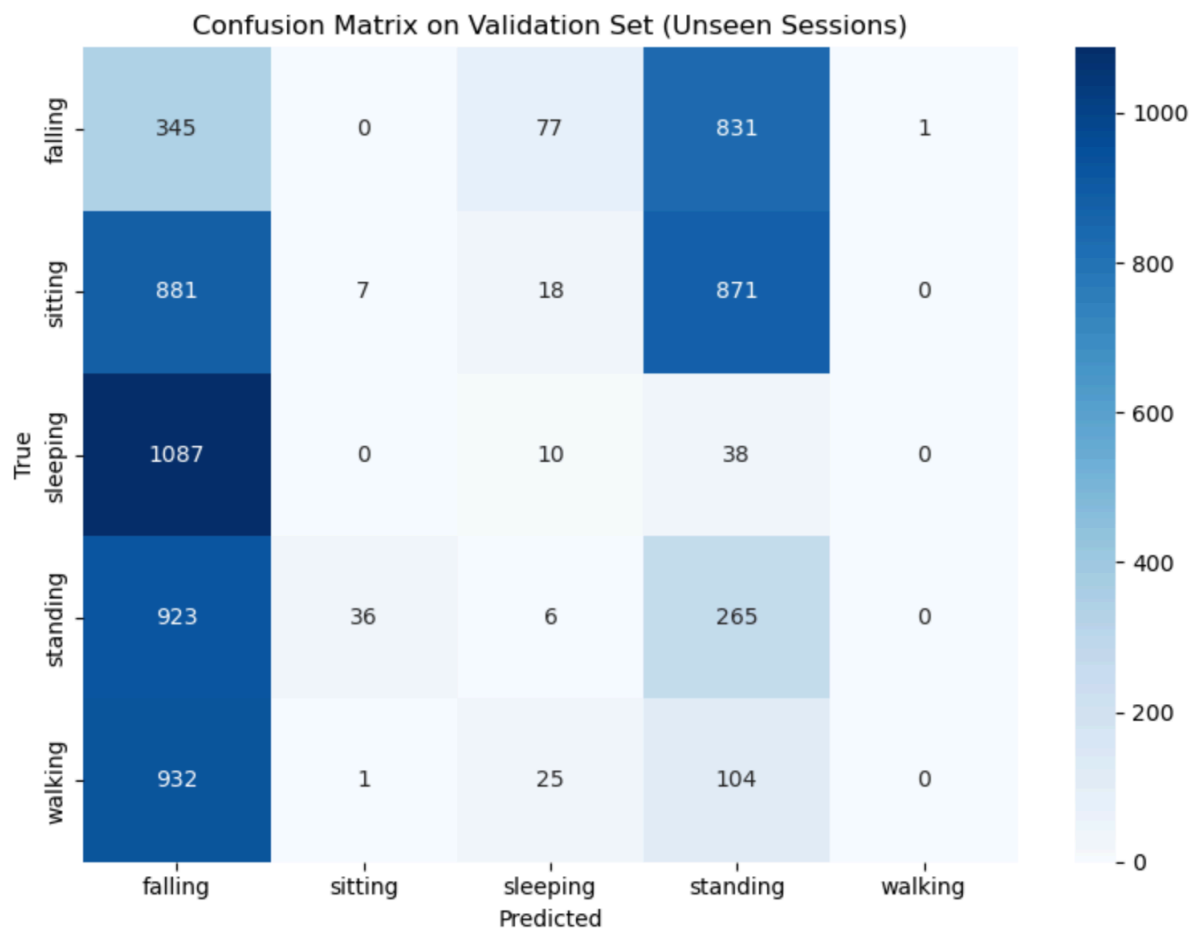
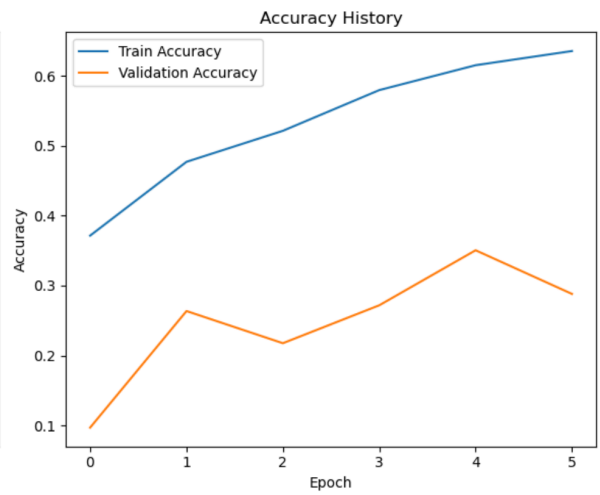
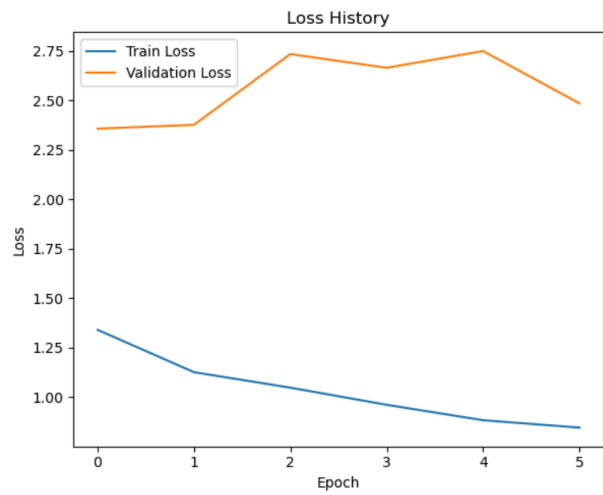
## Session 2 as Validation

```
train_sessions_files = [  
    'esp32_csi_walking_20250727_152811',  
    'esp32_csi_walking_20250727_152603',  
    'esp32_csi_walking_20250728_150535',  
    'esp32_csi_standing_20250727_155036',  
    'esp32_csi_standing_20250727_154720',  
    'esp32_csi_standing_20250728_141446',  
    'esp32_csi_sleeping_20250727_164127',  
    'esp32_csi_sleeping_20250727_163829',  
    'esp32_csi_sleeping_20250728_124410',  
    'esp32_csi_sitting_20250727_161517',  
    'esp32_csi_sitting_20250727_161310',  
    'esp32_csi_sitting_20250728_133443',  
    'esp32_csi_falling_20250727_133043',  
    'esp32_csi_falling_20250727_132826',  
    'esp32_csi_falling_20250728_114422'  
]
```

```
val_sessions_files = [  
    'esp32_csi_falling_20250727_165423',  
    'esp32_csi_sitting_20250728_125939',  
    'esp32_csi_sleeping_20250728_120223',  
    'esp32_csi_standing_20250728_134840',  
    'esp32_csi_walking_20250728_143754'  
]
```

### Training Logs and Results:

```
Starting model training...  
Epoch 1/50  
645/645 - 22s - 34ms/step - accuracy: 0.3716 - loss: 1.3403 - val_accuracy: 0.0971 - val_loss: 2.3574  
Epoch 2/50  
645/645 - 14s - 21ms/step - accuracy: 0.4770 - loss: 1.1263 - val_accuracy: 0.2637 - val_loss: 2.3770  
Epoch 3/50  
645/645 - 13s - 21ms/step - accuracy: 0.5214 - loss: 1.0480 - val_accuracy: 0.2177 - val_loss: 2.7346  
Epoch 4/50  
645/645 - 13s - 21ms/step - accuracy: 0.5795 - loss: 0.9613 - val_accuracy: 0.2719 - val_loss: 2.6652  
Epoch 5/50  
645/645 - 13s - 21ms/step - accuracy: 0.6151 - loss: 0.8834 - val_accuracy: 0.3506 - val_loss: 2.7499  
Epoch 6/50  
645/645 - 13s - 21ms/step - accuracy: 0.6354 - loss: 0.8462 - val_accuracy: 0.2882 - val_loss: 2.4856
```



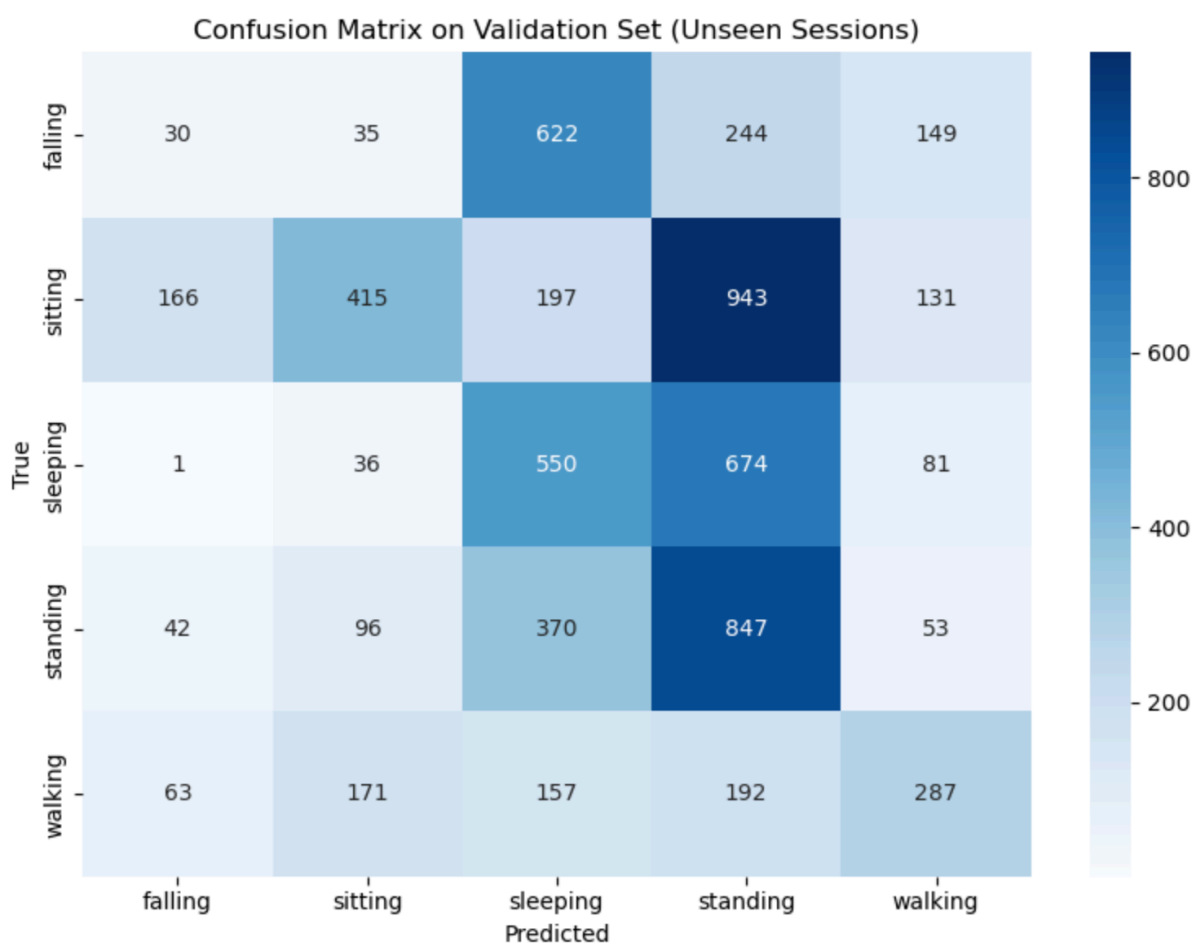
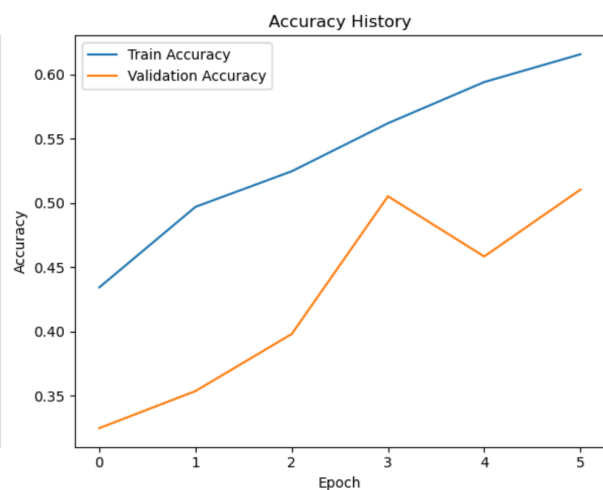
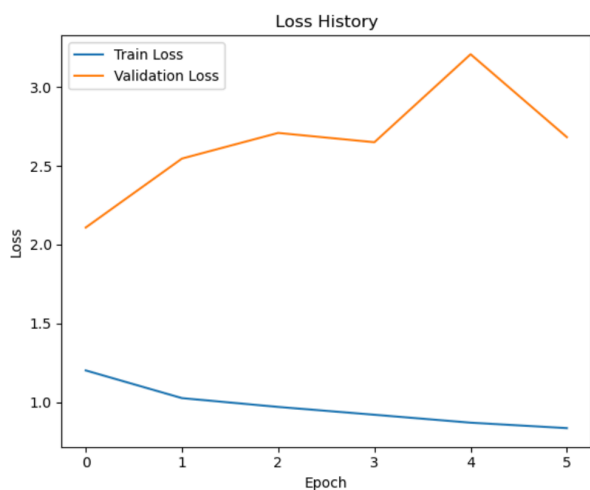
## Session 3 as Validation (Best)

```
train_sessions_files = [  
    'esp32_csi_walking_20250727_152811',  
    'esp32_csi_walking_20250727_152603',  
    'esp32_csi_walking_20250728_143955',  
    'esp32_csi_standing_20250727_155036',  
    'esp32_csi_standing_20250727_154516',  
    'esp32_csi_standing_20250728_135042',  
    'esp32_csi_sleeping_20250727_164127',  
    'esp32_csi_sleeping_20250727_163416',  
    'esp32_csi_sleeping_20250728_121235',  
    'esp32_csi_sitting_20250727_160929',  
    'esp32_csi_sitting_20250727_155903',  
    'esp32_csi_sitting_20250728_130356',  
    'esp32_csi_falling_20250727_132826',  
    'esp32_csi_falling_20250727_132002',  
    'esp32_csi_falling_20250727_165423'  
]
```

```
val_sessions_files = [  
    'esp32_csi_falling_20250728_114422',  
    'esp32_csi_sitting_20250728_132048',  
    'esp32_csi_sleeping_20250728_122928',  
    'esp32_csi_standing_20250728_142148',  
    'esp32_csi_walking_20250728_150750'  
]
```

### Training Logs and Results:

```
Starting model training...  
Epoch 1/50  
622/622 - 21s - 34ms/step - accuracy: 0.4345 - loss: 1.2032 - val_accuracy: 0.3249 - val_loss: 2.1088  
Epoch 2/50  
622/622 - 13s - 21ms/step - accuracy: 0.4971 - loss: 1.0277 - val_accuracy: 0.3538 - val_loss: 2.5458  
Epoch 3/50  
622/622 - 13s - 21ms/step - accuracy: 0.5248 - loss: 0.9719 - val_accuracy: 0.3982 - val_loss: 2.7082  
Epoch 4/50  
622/622 - 13s - 21ms/step - accuracy: 0.5622 - loss: 0.9223 - val_accuracy: 0.5053 - val_loss: 2.6493  
Epoch 5/50  
622/622 - 13s - 21ms/step - accuracy: 0.5942 - loss: 0.8721 - val_accuracy: 0.4585 - val_loss: 3.2068  
Epoch 6/50  
622/622 - 13s - 21ms/step - accuracy: 0.6158 - loss: 0.8376 - val_accuracy: 0.5105 - val_loss: 2.6817
```



# Analysis

Based on the experimental results, the CNN-LSTM model demonstrates moderate success, achieving 50-64% accuracy when trained and validated on data from the same individual. In these tests, it is good at identifying high-impact activities like falling but struggles to differentiate between static ones like sitting and standing.

However, the model's performance collapses to 10-51% accuracy in cross-session validation, indicating it fails to generalize to new individuals. The model learns features specific to an individual and their environment rather than general activity patterns, often developing a strong bias towards one or two classes when tested on unseen participant data.

I assume there is a data shift problem which demonstrates by the model overfitting to person-specific or environmental characteristics within the training data rather than learning the general patterns of the activities themselves.